BAYES AND KALMAN FILTER

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Abstract- This report discusses the working principles of the Bayes Filter, its mathematical components, the assumptions under which it reduces to the Kalman Filter, the equations involved in Kalman filtering, the significance of the Kalman Gain, and the application of these filtering techniques in Formula Driverless vehicles.

Keywords--Bayes Filter, Kalman Filter, State Estimation, Probability Distribution, Gaussian Noise, Sensor Fusion, Kalman Gain, Autonomous Vehicles.

I. INTRODUCTION

In autonomous vehicles, knowing the exact position and motion of the vehicle at every instant is essential for effective navigation and control. However, obtaining accurate information directly from sensors is challenging because every sensor has some degree of uncertainty. To overcome these uncertainties, state estimation techniques are used to combine information from mathematical motion models and sensor measurements. One such technique is the Bayes Filter, which uses probability theory to estimate the state of a system based on previous information, control inputs and new observations.

The Kalman Filter is a widely used implementation of Bayesian filtering. It estimates the state of a system by repeatedly predicting its future state and correcting that prediction using incoming sensor measurements. It is particularly useful in autonomous vehicles for combining sensor information and obtaining reliable estimates of vehicle position, velocity and orientation.

II. BAYES FILTER

The Bayes Filter is a probabilistic state-estimation method that estimates the current state of a dynamic system using previous state information, control inputs and sensor measurements. Instead of assuming that the system's state is known exactly, it represents the state using a probability distribution called the **belief state**.

The belief state is mathematically represented as:

$$\text{bel}(x_t) = P(x_t|z_1, u_1)$$

The Bayes Filter works through two main stages: prediction and measurement update.

III. PRIOR BELIEF

The prior belief represents the available knowledge about the system's state before incorporating the latest sensor measurement. It is based on the previous state estimate and all the information collected up to the previous time step.

Mathematically, the previous belief is represented as:

$$\text{bel}(x_{t-1}) = P(x_{t-1}|\text{z}_1, \text{u}_1)$$

IV. MOTION MODEL

The motion model describes how the system is expected to change when a control input is applied. It is represented by the conditional probability:

$$P(x_t \mid x_{t-1}, u_t)$$

V. MEASUREMENT LIKELIHOOD

The measurement likelihood represents the probability of obtaining a particular sensor measurement if the system is in a particular state. It describes how well a possible state explains the actual observation received from the sensor.

$$P(z_t \mid x_t)$$

VI. PREDICTION AND UPDATE

The prediction step estimates the next state using the previous belief and the system's motion model.

$$\overline{\text{bel}}(x_t) = \int P\,(x_t \mid x_{t-1}, u_t)\text{bel}(x_{t-1})\,dx_{t-1}$$

The measurement update step corrects the predicted belief using the latest sensor observation. It combines the predicted belief with the measurement likelihood to produce an updated belief.

$$\text{bel}(x_t) = \eta P(z_t \mid x_t)\overline{\text{bel}}(x_t)$$

VII. NORMALISATION

After the measurement update, the resulting probability distribution must be normalised so that the total probability equals 1. The normalisation constant ensures that the updated

belief forms a valid probability distribution.

$$\eta = \frac{1}{\int P\,(z_t \mid x_t)\overline{\text{bel}}(x_t)\,dx_t}$$

The denominator represents the total probability of obtaining the observed measurement across all possible states. The normalisation constant scales the updated probabilities so that their total equals 1.

VIII. KALMAN FILTER

The Kalman Filter is a specialised form of the Bayes Filter used for estimating the state of a dynamic system. It is particularly suitable for systems in which the state transition and measurement models are linear and the uncertainties follow Gaussian distributions.

Unlike a general Bayes Filter, which may require calculations involving entire probability distributions, the Kalman Filter represents the state belief using only two quantities: the **mean and covariance**. The mean represents the estimated state, while the covariance describes the uncertainty associated with that estimate.

IX. MATHEMATICAL STAGES OF KALMAN FILTER

1. *State prediction*

   The first step predicts the current state using the previous state estimate and the control input.

$$\hat{x}_t^- = F\hat{x}_{t-1} + Bu_t$$

2. *Covariance prediction*

   The second step calculates the uncertainty associated with the predicted state. Process noise represents uncertainties in the system's motion, such as unexpected changes in acceleration or inaccuracies in the motion model. The covariance prediction accounts for how uncertainty changes as the system evolves.

$$P_t^- = FP_{t-1}F^T + Q$$

   Covariance matrix :

$$\mathbf{P} = \begin{bmatrix} \sigma_p^2 & \sigma_{pv} \\ \sigma_{pv} & \sigma_v^2 \end{bmatrix}$$

3. *Kalman gain calculation*

   The Kalman Gain determines how much importance should be given to the predicted state compared with the new sensor measurement. It controls the extent to which the predicted estimate is corrected.

   First, the innovation or measurement residual is calculated:

$$y_t = z_t - H\hat{x}_t^-$$

   The innovation covariance is:

$$S_t = HP_t^-H^T + R$$

   The Kalman Gain is then calculated using:

$$K_t = P_t^-H^TS_t^{-1}$$

   Innovation represents the difference between the actual sensor measurement and the predicted measurement. The Kalman Gain determines how strongly the filter responds to this difference.

4. *State update / correction*

   After calculating the Kalman Gain, the predicted state is corrected using the difference between the actual measurement and the predicted measurement.

$$\hat{x}_t = \hat{x}_t^- + K_t(z_t - H\hat{x}_t^-)$$

5. *Covariance update*

   The final step updates the covariance to reflect the reduced uncertainty after incorporating the new measurement.

$$P_t = (I - K_tH)P_t^-$$

   Here, (I) is the identity matrix. This equation updates the uncertainty associated with the corrected state estimate. Once the state and covariance have been updated, they are used as the starting point for the next prediction cycle.

X. ASSUMPTIONS UNDER WHICH BAYES FILTER REDUCED TO KALMAN FILTER

Linear State Transition Model

The system is assumed to follow a linear state transition model, meaning that the current state can be represented as a linear function of the previous state and control input. It is expressed as:

$$x_t = Fx_{t-1} + Bu_t + w_t$$

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

Here, (F) is the state-transition matrix, (B) is the control-input matrix, (u_t) is the control input, and (w_t) represents process noise.

Linear Measurement Model

The sensor measurements are assumed to be linearly related to the actual state of the system.

$$z_t = Hx_t + v_t$$

Gaussian Process Noise

The uncertainty in the system's motion, called process noise, is assumed to follow a Gaussian or normal distribution. It represents unpredictable variations in the system, such as changes in vehicle motion due to road conditions or inaccuracies in the motion model. Process noise is generally assumed to have zero mean and a covariance matrix (Q). The covariance matrix describes the degree of uncertainty associated with the system's motion. This assumption allows the Kalman Filter to mathematically account for motion uncertainty during the prediction stage.

Gaussian Initial Belief

The initial belief about the system's state is assumed to follow a Gaussian distribution. This means that the initial state uncertainty can be represented using a mean and a covariance matrix. The mean represents the estimated state, while the covariance describes the uncertainty associated with that estimate. This assumption is important because the Kalman Filter maintains a Gaussian belief throughout its prediction and correction stages, provided the other assumptions are also satisfied.

XI. WHEN THESE ASSUMPTIONS DO NOT HOLD TRUE

If the system's motion involves nonlinear relationships, the standard Kalman Filter may not accurately predict the next state. An Extended Kalman Filter (EKF) can be used to approximate nonlinear functions through local linearisation**.**

Some sensors provide measurements that are related to the system's state through nonlinear relationships. In such cases, the standard Kalman Filter may not process the measurements accurately. EKF or Unscented Kalman Filter (UKF) techniques can be used to handle nonlinear measurement models.

If the noise follows a distribution significantly different from a Gaussian distribution, representing uncertainty using only the mean and covariance may not be sufficient. For example, sensor measurements may occasionally contain large errors or outliers. Particle Filters and robust filtering techniques may be more suitable for such situations.

If process noise, measurement noise or the initial state are correlated, the standard covariance calculations may no longer be valid without modification. Estimation models that account for these correlations may be required.

XII. KALMAN GAIN

The Kalman Gain plays an important role in determining how much the filter trusts the predicted state compared with the sensor measurement. Its value depends on the uncertainty in the prediction and the uncertainty in the measurement.

$$K = \frac{P^-}{P^- + R}$$

Here, (P^-) represents the predicted-state variance and (R) represents the measurement-noise variance. When the sensor is highly accurate, its measurement uncertainty (R) is small. Therefore, the Kalman Gain becomes larger, and the filter gives more importance to the sensor measurement. On the other hand, when the sensor is noisy, (R) becomes larger, resulting in a smaller Kalman Gain. The filter then relies more heavily on its predicted state.

If the predicted state has high uncertainty, the filter gives more importance to the incoming measurement. If the prediction is highly reliable, the correction made using the measurement is smaller.

XIII. APPLICATION IN DIVERLESS CAR

In a Formula Driverless vehicle, sensors such as GPS/GNSS, IMU, LiDAR and wheel-speed sensors provide different types of information These measurements may contain noise and uncertainty. A filtering algorithm combines the sensor observations with the vehicle's motion model to estimate its current state more accurately.

The Kalman Filter fits into the state-estimation or sensor-fusion stage of the autonomous driving pipeline. The Kalman Filter is positioned between sensor processing and higher-level planning and control. Its output provides reliable information about the vehicle's current state, allowing the remaining autonomous driving modules to make informed decisions. Its main purpose is to provide reliable information about the vehicle's current state, which is then used by other modules of the autonomous driving system. Since vehicle motion can involve nonlinear relationships, an Extended Kalman Filter or Unscented Kalman Filter may be more appropriate than a standard Kalman Filter in certain vehicle-state estimation applications.

XIV. CONCLUSION

The Bayes Filter provides a mathematical framework for estimating the state of a dynamic system by combining predictions from a motion model with observations obtained from sensors. It operates through prediction and measurement-update stages, using prior belief, measurement likelihood and normalisation to calculate an updated probability distribution. The Kalman Filter is a specialised form of Bayesian filtering that simplifies these calculations under linear-model and Gaussian-noise assumptions. It operates through state prediction, covariance prediction, Kalman Gain calculation, state correction and covariance update.

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XVI. REFERENCES

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